HOVID LIPIDUCE

VILIPxx-0

DESCRIPTION
Lipiduce-10
White coloured film-coated oval shaped tablets, shallow convex faces, plain on both sides.

Lipiduce-20
White coloured film-coated oval shaped tablets, shallow convex with "HD" and break bar embossed on one face.

COMPOSITION
Each film-coated tablet contains:
Lipiduce-10
Atorvastatin Calcium equivalent to Atorvastatin 10 mg.

Lipiduce-20
Atorvastatin Calcium equivalent to Atorvastatin 20 mg.

PHARMACODYNAMICS
Atorvastatin is a selective, competitive inhibitor of 3-Hydroxy-3-methylglutaryl coenzyme A (HMG-CoA) reductase. The inhibition of HMG-CoA reductase prevents conversion of HMG-CoA to mevalonate, the rate-limiting step in cholesterol biosynthesis. Inhibition of cholesterol synthesis in the liver leads to upregulation of low-density lipoprotein (LDL) receptors and an increase in catabolism of LDL cholesterol. It may also reduce the production of LDL as a result of inhibition of hepatic synthesis of very low-density lipoprotein (VLDL), the precursor of LDL.

HMG-CoA reductase inhibitors reduce LDL-cholesterol, VLDL-cholesterol, and to a lesser extent, plasma triglyceride concentrations, and slightly increase high-density lipoprotein (HDL) concentration.

PHARMACOKINETICS
Atorvastatin is rapidly absorbed after oral administration; maximum plasma concentration occurs within 1 to 2 hours. It has low absolute bioavailability of about 12% due to presystemic clearance in the gastrointestinal mucosa and / or first-pass metabolism in the liver, its primary sites of action.

Atorvastatin is 98% bound to plasma proteins. It is metabolized by ortho- and parahydroxylation and beta- oxidation to active metabolites. The drug and its metabolites are eliminated primarily in the bile. The mean plasma elimination half-life is approximately 14 hours.

INDICATIONS
Atorvastatin is indicated as an adjunct to diet for the treatment of patients with elevated total cholesterol (total-C), low density lipoprotein cholesterol (LDL-C), apolipoprotein B (apo B), and triglycerides (TG) and to increase high density lipoprotein cholesterol (HDL-C) in patients with primary hypercholesterolemia (heterozygous familial and nonfamilial hypercholesterolemia), combined (mixed) hyperlipidemia (*Fredrickson* Types IIa and IIb), elevated serum TG levels (*Fredrickson* Type IV), and for patients with dysbetalipoproteinemia (*Fredrickson* Type III) who do not respond adequately to diet.

Atorvastatin is also indicated for the reduction of total-C and LDL-C in patients with homozygous familial hypercholesterolemia.

Prevention of Cardiovascular Disease
In adult patients without clinically evident cardiovascular disease (CVD), but with multiple risk factors for coronary heart disease (CHD) such as age, smoking, hypertension, low HDL-C, or a family history of early CHD, LIPIDUCE is indicated to:
- Reduce the risk of myocardial infarction (MI)
- Reduce the risk of stroke
- Reduce the risk for revascularization procedures and angina

In patients with type 2 diabetes, and without clinically evident CHD, but with multiple risk factors for coronary heart disease such as retinopathy, albuminuria, smoking, or hypertension, LIPIDUCE is indicated to:
- Reduce the risk of myocardial infarction
- Reduce the risk of stroke

In patients with clinically evident CHD, atorvastatin is indicated to:
- Reduce the risk of non-fatal MI
- Reduce the risk of fatal and non-fatal stroke
- Reduce the risk for revascularization procedures
- Reduce the risk of hospitalization for congestive heart failure (CHF)
- Reduce the risk of angina

Pediatric Patients (10-17 years of age)
Atorvastatin is indicated as an adjunct to diet to reduce total-C, LDL-C, and apo B levels in boys and postmenarchal girls, 10 to 17 years of age, with heterozygous familial hypercholesterolemia if after an adequate trial of diet therapy the following findings are present:
- LDL-C remains ≥190 mg/dL or
- LDL-C remains ≥160 mg/dL and
  - There is a positive family history of premature CVD or
  - Two or more other CVD risk factors are present in the pediatric patient

CONTRAINDICATIONS
- Active liver disease or unexplained persistent elevations of serum transaminases.
- Hypersensitivity to any component of this drug
- Pregnancy and lactation.

PRECAUTIONS
- Skeletal Muscle Effects: Physicians considering combined therapy with atorvastatin and fibrates, erythromycin, immunosuppressive drugs, azole antifungals, or lipid-modifying doses of niacin (≥ 1g/day) should carefully weigh the potential benefits and risks and should carefully monitor patients for any signs and symptoms of muscle pain, tenderness, or weakness, particularly during the initial months of therapy and during any periods of upward dosage titration of either drug. Therefore, lower starting and maintenance doses of atorvastatin should also be considered when taken concomitantly with the aforementioned drugs. Temporary suspension of atorvastatin may be appropriate during fusidic acid therapy.
- There have been very rare reports of an immune-mediated necrotizing myopathy (IMNM) during or after treatment with some statins. IMNM is clinically characterized by:
  - persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment;
  - muscle biopsy showing necrotizing myopathy without significant inflammation;
  - improvement with immunosuppressive agents.
- Myasthenia Gravis/Ocular Myasthenia: In few cases, statins have been reported to induce de novo or aggravate pre-existing myasthenia gravis or ocular myasthenia. Lipiduce tablet should be discontinued in case of aggravation of symptoms. Recurrences when the same or a different statin was (re-) administered have been reported.
- Liver Dysfunction: HMG-CoA reductase inhibitors have been associated with biochemical abnormalities of liver function. Therefore, liver function tests should be performed prior to and at 12 weeks following initiation of therapy or elevation in dose, and semiannually thereafter.
- Atorvastatin should be used with caution in patients with alcoholism, organ transplant with immunosuppressant therapy and/or patients who have a history of liver disease.
- Serious conditions such as hypotension, severe acute infection, severe metabolic, endocrine, or electrolyte disorder, uncontrolled seizures, major surgery, or trauma may increased risk of secondary renal failure if rhabdomyolysis occurs.

PREGNANCY AND LACTATION
Atorvastatin is not recommended during pregnancy, in women who plan to become pregnant in the near future, or during lactation.

MAIN SIDE/ADVERSE EFFECTS
Myalgia, myositis, rhabdomyolysis, constipation, diarrhea, heartburn, stomach pain, dizziness, headache, nausea, skin rash, insomnia.

Musculoskeletal disorders
Frequency not known:
Immune-mediated necrotizing myopathy

There have been rare post-marketing reports of cognitive impairment (e.g. memory loss, forgetfulness, amnesia, memory impairment, confusion) associated with statin use. These cognitive issues have been reported for all statins. The reports are generally non-serious and reversible upon statin discontinuation, with variable times to symptom onset (1 day to years) and symptom resolution (median 3 weeks).

Increases in HbA1c and fasting blood glucose have been reported with statins. The risk of hyperglycemia, however, is outweighed by the reduction in vascular risk with statins.

Nervous system disorders
Frequency not known:
Myasthenia gravis

Eye disorders
Frequency not known:
Ocular myasthenia

DRUG INTERACTIONS
The risk of myopathy during treatment with HMG-CoA reductase inhibitors is increased with concurrent administration of cyclosporine, fibric acid derivatives, niacin or cytochrome P450 3A4 inhibitors (eg erythromycin and azole antifungals).

Inhibitors of cytochrome P450 3A4: Atorvastatin is metabolized by cytochrome P450 3A4. Concomitant administration of atorvastatin with inhibitors of cytochrome P450 3A4 can lead to increases in plasma concentrations of atorvastatin. The extent of interaction and potentiation of effects depends on the variability of effect on cytochrome P450 3A4.

Transporter Inhibitors: Atorvastatin and atorvastatin- metabolites are substrates of the OATP1B1 transporter. Inhibitors of the OATP1B1 (e.g. cyclosporine) can increase the bioavailability of atorvastatin. Concomitant administration of atorvastatin 10 mg and cyclosporine 5.2 mg/kg/day resulted in a 7.7 fold increase in exposure to atorvastatin.

Erythromycin/Clarithromycin: Co-administration of atorvastatin and erythromycin (500 mg four times daily), or clarithromycin (500 mg twice daily) known inhibitors of cytochrome P450 3A4, was associated with higher plasma concentrations of atorvastatin.
Protease inhibitors: Co-administration of atorvastatin and protease inhibitors, known inhibitors of cytochrome P450 3A4, was associated

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with increased plasma concentrations of atorvastatin.

Diltiazem hydrochloride: Co-administration of atorvastatin (40mg) with diltiazem (240mg) was associated with higher plasma concentrations of atorvastatin.

Cimetidine: No clinically significant interactions were seen.

Itraconazole: Concomitant administration of atorvastatin (20 to 40 mg) and itraconazole (200 mg) was associated with an increase in atorvastatin AUC.

Grapefruit juice: Contains one or more components that inhibit CYP 3A4 and can increase plasma concentrations of atorvastatin, especially with excessive grapefruit juice consumption (>1.2 liters per day).

Inducers of cytochrome P450 3A: Concomitant administration of atorvastatin with inducers of cytochrome P450 3A4 (eg efavirenz, rifampin) can lead to variable reductions in plasma concentrations of atorvastatin. Due to the dual interaction mechanism of rifampin, (cytochrome P450 3A4 induction and inhibition of hepatocyte uptake transporter OATP1B1), simultaneous co-administration of atorvastatin with rifampin is recommended, as delayed administration of atorvastatin after administration of rifampin has been associated with a significant reduction in atorvastatin plasma concentrations.

Antacids: Co-administration of atorvastatin with an oral antacid suspension containing magnesium and aluminum hydroxides, decreased atorvastatin plasma concentrations approximately 35%; however, LDL-C reduction was not altered.

Antipyrine: Because atorvastatin does not affect the pharmacokinetics of antipyrine, interactions with other drugs metabolized via the same cytochrome isozymes are not expected.

Colestipol: Plasma concentrations of atorvastatin were lower (approximately 25%) when colestipol was administered with atorvastatin. However, lipid effects were greater when atorvastatin and colestipol were co-administered than when either drug was given alone.

Digoxin: When multiple doses of digoxin and 10 mg atorvastatin were co-administered, steady- state plasma digoxin concentrations were unaffected. However, digoxin concentrations increased approximately 20% following administration of digoxin with 80 mg atorvastatin daily. Patients taking digoxin should be monitored appropriately.

Azithromycin: Co-administration of atorvastatin (10 mg once daily) and azithromycin (500 mg once daily) did not alter the plasma concentrations of atorvastatin.

Oral Contraceptives: Co-administration with an oral contraceptive containing norethindrone and ethinyl estradiol increased AUC values for norethindrone and ethinyl estradiol by approximately 30% and 20%. These increases should be considered when selecting an oral contraceptive for a woman taking atorvastatin.

Warfarin: Co-administration of atorvastatin 80mg daily with warfarin will cause small decrease in prothrombin time and increase bleeding time. Although only very rare cases of clinically significant anticoagulant interactions have been reported, prothrombin time should be determined before starting atorvastatin in patients taking coumarin anticoagulants and frequently enough during early therapy to ensure that no significant alteration of prothrombin time occurs.

Amlodipine: Co-administration of atorvastatin 80 mg and amlodipine 10 mg resulted in an 18% increase in exposure to atorvastatin which was not clinically meaningful.

Colchicine: Although interaction studies with atorvastatin and colchicine have not been conducted, cases of myopathy have been reported with atorvastatin co-administered with colchicine, and caution should be exercised when prescribing atorvastatin with colchicine.

Fusidic Acid: Although interaction studies with atorvastatin and fusidic acid have not been conducted, severe muscle problems such as rhabdomyolysis have been reported in post-marketing experience with this combination. Patients should be closely monitored and temporary suspension of atorvastatin treatment may be appropriate.

Fibrates: Concurrent use of fibrates may cause severe myositis and myoglobinuria.

Other Concomitant Therapy: Atorvastatin was used concomitantly with antihypertensive agents and estrogen replacement therapy without evidence of clinically significant adverse interactions. Interaction studies with specific agents have not been conducted.

OVERDOSE

There is no specific treatment of atorvastatin overdosage. In the event of an overdose, the patient should be treated symptomatically, and supportive measures instituted as required.

DOSAGE AND ADMINISTRATION

General - Before instituting therapy with atorvastatin, an attempt should be made to control hypercholesterolemia with appropriate diet, exercise and weight reduction in obese patients, and to treat the underlying medical problems. The patient should continue on a standard cholesterol-lowering diet during treatment with atorvastatin. The dosage range is 10 mg to 80 mg once daily. Doses may be given any time of the day, with or without food. Starting and maintenance dosage should be individualized according to baseline LDL-C levels, the goal of therapy, and patient response. After initiation and/or upon titration of atorvastatin, lipid levels should be analyzed within 2 to 4 weeks, and dosage adjusted accordingly.

Primary Hypercholesterolemia and Combined (Mixed) Hyperlipidemia

The majority of patients are controlled with 10 mg atorvastatin once daily. A therapeutic response is evident within 2 weeks, and the maximum response is usually achieved within 4 weeks. The response is maintained during chronic therapy.

Homozygous Familial Hypercholesterolemia

In a compassionate-use study of patients with homozygous familial hypercholesterolemia, most patients responded to 80 mg atorvastatin with a greater than 15% reduction in LDL-C (18%-45%).

Heterozygous Familial Hypercholesterolemia in Pediatric Patients (10-17 years of age)

The recommended starting dose of atorvastatin is 10 mg/day; the usual dose range is 10 to 20 mg orally once daily. Doses should be individualized according to the recommended goal of therapy. Adjustments should be made at intervals of 4 weeks or more.

Severe Dyslipidemias in Pediatric Patients

For patients aged 10 years and above, the recommended starting dose is 10 mg atorvastatin once daily. The dose may be increased to 80 mg daily, according to the response and tolerability. Doses should be individualized according to the recommended goal of therapy. Adjustments should be made at intervals of 4 weeks or more. Experience in pediatric patients younger than 10 years of age is derived from open-label studies.

Use in Patients with Hepatic Insufficiency

Refer to “Contraindication” and “Precautions”.

Use in Patients with Renal Insufficiency

Renal disease has no influence on plasma concentrations or on LDL-C reduction with atorvastatin. Thus, no dose adjustment is required.

Use in the Elderly

No differences in safety, efficacy or lipid treatment goal attainment were observed between elderly patients and the overall population

Dosage in Patients Taking Cyclosporine, Clarithromycin, Itraconazole, or Certain Protease Inhibitors

In patients taking cyclosporine or the HIV protease inhibitors (tipranavir plus ritonavir) or the hepatitis C protease inhibitor (telaprevir), therapy with Lipiduce should be avoided.

In patients with HIV taking lopinavir plus ritonavir, caution should be used when prescribing Lipiduce and the lowest dose necessary is employed.

In patients taking clarithromycin, itraconazole, or in patients with HIV taking a combination of saquinavir plus ritonavir, darunavir plus ritonavir, fosamprenavir, or fosamprenavir plus ritonavir, therapy with Lipiduce should be limited to 20 mg, and appropriate clinical assessment is recommended to ensure that the lowest dose necessary of Lipiduce is employed.

In patients taking the HIV protease inhibitor nelfinavir or the hepatitis C protease inhibitor boceprevir, therapy with Lipiduce should be limited to 40 mg, and appropriate clinical assessment is recommended to ensure that the lowest dose necessary of Lipiduce is employed.

Use in Children (Homozygous Familial Hypercholesterolemia)

Treatment experience in a pediatric population is limited to doses of atorvastatin up to 80 mg/day for one year in 8 patients with homozygous FH. No clinical or biochemical abnormalities were reported in these patients.

Note: The information given here is limited. For further information, consult your doctor or pharmacist.

Storage: Store below 30°C. Protect from light and moisture.

Presentation/Packing: Alu-alu blister of 3 x 10's, 10 x 10's.

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